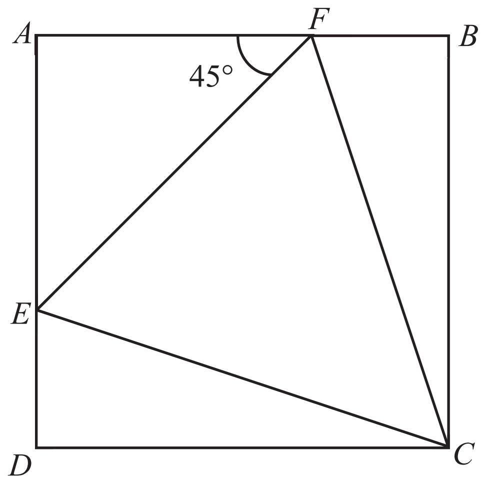
21 In the figure, *ABCD* is a square, $\angle AFE = 45°$ and the ratio of *AF* : *FB* is 2 : 1.



(a) Name a pair of congruent triangles and state the reasons for the congruency.

Answer ………………………………………………………………………………… [2]

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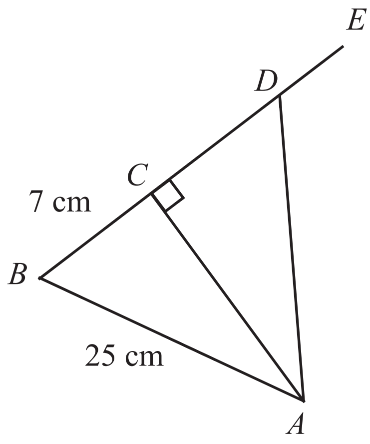
(b) Find the ratio of the area of ΔCEF to the area of square *ABCD*.

Answer …………………………….. [2]

(c) Given that the area of ΔCEF is 16 cm^2. Find the length of the side of the square *ABCD*.

Answer ………………………… cm [2]

22 In the diagram, $BCDE$ is a straight line, $AB = 25$ cm, $BC = 7$ cm and $\angle ACD = 90°$.



Given that $\tan \angle CAD = \dfrac{3}{4}$, find

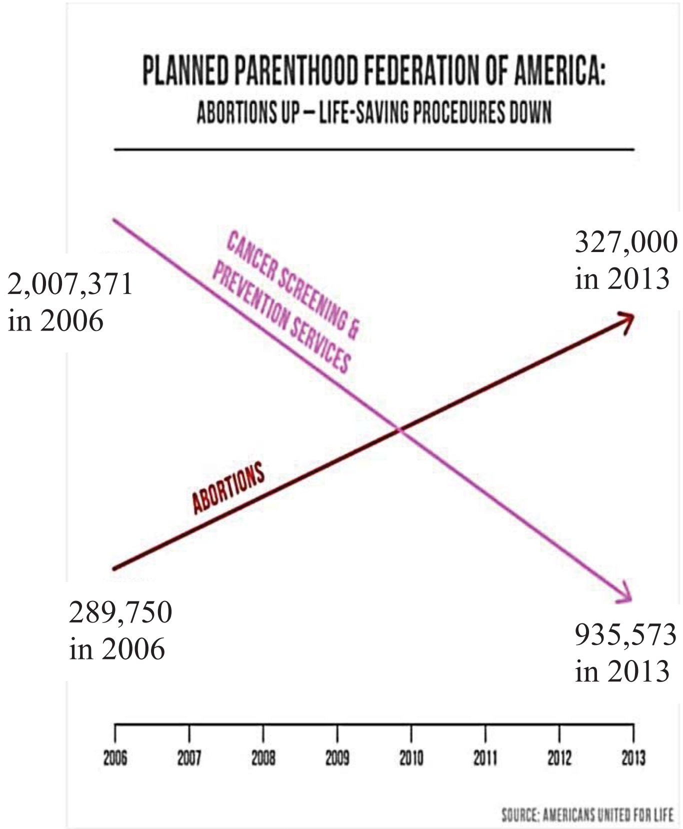
(a) the length of CD,

Answer $CD =$ …………………. cm [2]

(b) $\cos \angle ADE$.

Answer $\cos \angle ADE =$ ……………..... [2]

23 The chart below shows the annual number of Americans who went for cancer screening and abortion.



(a) Use the graph to calculate the annual rate of decrease of cancer screening from 2006 and 2013.

Answer …………………………….. [1]

(b) Explain why this graph is misleading.

Answer ……………………………………………………………………………

……………………………………………………………………………

…………………………………………………………………………… [1]